

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12416736
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Aoki; Toru et al.

Radiographic device

Abstract

A radiation imaging apparatus includes a plurality of pixels configured to generate an electric charge corresponding to energy or the number of particles of incident radiation, a plurality of signal processing parts respectively connected to the plurality of pixels and configured to generate a digital value based on the electric charge provided by each of the pixels, and a circuit board in which a radiation detector including the plurality of pixels and the plurality of signal processing parts is disposed in a two-dimensional manner. Each of the plurality of signal processing parts includes a signal conversion part configured to convert an analog value based on the electric charge into the digital value, and a clock signal generation part configured to provide a clock signal for generating a digital value to the signal conversion part.

Inventors:	Aoki; Toru (Hamamatsu, JP), Takagi; Katsuyuki (Hamamatsu, JP), Koike; Akifumi (Hamamatsu, JP)
Applicant:	NATIONAL UNIVERSITY CORPORATION SHIZUOKA UNIVERSITY (Shizuoka, JP); ANSeeN Inc. (Shizuoka, JP)
Family ID:	1000008819957
Assignee:	NATIONAL UNIVERSITY CORPORATION SHIZUOKA UNIVERSITY (N/A, JP); ANSeeN Inc. (N/A, JP)
Appl. No.:	18/252128
Filed (or PCT Filed):	November 13, 2020
PCT No.:	PCT/JP2020/042488
PCT Pub. No.:	WO2022/102095
PCT Pub. Date:	May 19, 2022

Prior Publication Data

Publication Classification**Int. Cl.:** G01T1/24 (20060101)**U.S. Cl.:**

CPC G01T1/247 (20130101);

Field of Classification Search**USPC:** None

References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2014/0226166	12/2013	Kumar	356/601	H04N 25/78

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
108618792	12/2017	CN	N/A
111919140	12/2019	CN	N/A
S59-137874	12/1983	JP	N/A
2010-021151	12/2009	JP	N/A
2011-085479	12/2010	JP	N/A
2012-032322	12/2011	JP	N/A
2017-096798	12/2016	JP	N/A
2020-507753	12/2019	JP	N/A
WO 2018/144127	12/2017	WO	N/A
WO 2020/077217	12/2019	WO	N/A

OTHER PUBLICATIONS

International Search Report mailed Jan. 26, 2021 in corresponding PCT International Application No. PCT/JP2020/042488. cited by applicant

English translation of the International Preliminary Report on Patentability (IPRP) (Chapter 1 or II of the PCT Treaty) mailed May 25, 2023 with a Notification from the International Bureau (Form PCT/IB/338) issued in corresponding International Application No. PCT/JP2020/042488. cited by applicant

Primary Examiner: Gunberg; Edwin C

Attorney, Agent or Firm: OSTROLENK FABER LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application is a U.S. national stage application of PCT/JP2020/042488, filed Nov. 13, 2020, which is incorporated by this reference.

TECHNICAL FIELD

(2) The present invention relates to a radiation imaging apparatus.

BACKGROUND ART

(3) A technology for detecting radiation has been developed. The radiation detection technology is expected to be applied to the medical field, industrial field, security field, and the like. As an apparatus for detecting radiation, there is a radiation imaging apparatus for obtaining a radiation image. The radiation imaging apparatus outputs image data that indicates a state of incident radiation. To generate the image data, for example, signal processing such as converting an analog signal into a digital signal is performed. The signal processing is performed by providing a clock signal for driving an electronic circuit to the electronic circuit that exhibits a desired function. Patent Literature 1 and Patent Literature 2 disclose apparatuses for detecting X-rays or radiation. Further, Patent Literature 1 and Patent Literature 2 show that the apparatuses operate in response to a clock.

CITATION LIST

Patent Literature

(4) [Patent Literature 1] Japanese Unexamined Patent Publication No. 2020-507753 [Patent Literature 2] PCT International Publication No. WO2020-077217

SUMMARY OF INVENTION

Technical Problem

(5) In the technical field of radiation imaging apparatuses, it is desired to increase an area of an effective pixel region. The expansion of the effective pixel region is accompanied by an increase in the number of pixels. As the number of pixels increases, power consumption also tends to increase.

(6) An object of the present invention is to provide a radiation imaging apparatus capable of curbing increase in power consumption.

Solution to Problem

(7) A radiation imaging apparatus according to an embodiment of the present invention includes a plurality of pixels configured to generate electric charge corresponding to energy or the number of particles of incident radiation, a plurality of signal processing parts respectively connected to the plurality of pixels and configured to generate a digital value based on the electric charge provided by each of the pixels, and a circuit board in which a radiation detector including the plurality of pixels and the plurality of signal processing parts is disposed in a two-dimensional manner. Each of the plurality of signal processing parts includes a signal conversion part configured to convert an analog value based on the electric charge into the digital value, and a clock signal generation part configured to provide a clock signal for generating a digital value to the signal conversion part.

(8) In the radiation imaging apparatus, the signal processing part is provided for each of the pixels. The signal processing part includes the signal conversion part and the clock signal generation part. That is, in the radiation imaging apparatus, the clock signal generation part is provided for each of the pixels. With such a configuration, it is possible to drive only the signal conversion part corresponding to the pixel on which the radiation is incident and not to drive the signal conversion part corresponding to a pixel on which radiation is not incident. Therefore, even when the number of pixels is increased, the signal conversion parts which do not need to be operated are not operated, and thus increase in power consumption can be curbed.

(9) The clock signal generation part may start an operation of providing the clock signal to the signal conversion part triggered by generation of the electric charge due to incidence of the radiation on the pixels. With such a configuration, the signal conversion part can be operated each time radiation is incident.

(10) The radiation imaging apparatus according to the embodiment of the present invention may further include a transmission signal generation part configured to provide a transmission signal for outputting the digital value from the signal processing part to the circuit board to each of the plurality of signal processing parts, and a wiring unit configured to transmit the transmission signal to the signal processing part. With such a configuration, the output operations of the plurality of signal processing parts can be synchronized.

Advantageous Effects of Invention

(11) According to the present invention, there is provided a radiation imaging apparatus capable of suppressing an increase in power consumption.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a perspective view showing a configuration of a radiation imaging apparatus according to an embodiment.

(2) FIG. 2 is a perspective view of the radiation imaging apparatus shown in FIG. 1 when seen from a first viewpoint in a state in which the radiation imaging apparatus is disassembled.

(3) FIG. 3 is a perspective view of the radiation imaging apparatus shown in FIG. 1 when seen from a second viewpoint in the state in which the radiation imaging apparatus is disassembled.

(4) FIG. 4 is an enlarged view showing a radiation detector included in the radiation imaging apparatus of FIG. 1.

(5) FIG. 5 is a block diagram showing a configuration of a signal processing part included in the radiation imaging apparatus shown in FIG. 4.

(6) FIG. 6 is a diagram for describing an operation of the radiation imaging apparatus.

DESCRIPTION OF EMBODIMENTS

(7) A radiation imaging apparatus **1** shown in FIG. 1 obtains a two-dimensional image based on radiation arriving from an inspection target. The radiation is, for example, gamma rays, X-rays, alpha rays, beta rays, and the like. The radiation imaging apparatus **1** includes a plurality of radiation detectors **2** and a circuit board **3**. The radiation detectors **2** are mounted on a circuit main surface **3a** of the circuit board **3**. Each of the radiation detectors **2** outputs a pixel signal corresponding to incident radiation. The radiation detector **2** includes a plurality of pixels **G** (refer to FIG. 4) and outputs a pixel signal for each of the pixels **G**. The pixel signal is used to generate a radiation image in an image generation part **4**. The circuit board **3** receives the pixel signal output from the radiation detector **2**. The circuit board **3** has a signal line for providing a control signal from a control part **5** to the radiation detector **2**. Further, the circuit board **3** has a signal line for outputting a pixel signal to the image generation part **4**. Further, the circuit board **3** may have a signal line for electrically connecting the radiation detectors **2** to each other.

(8) As shown in FIGS. 2 and 3, the radiation detector **2** includes an electric charge generation part **10** and a reading part **20**. Shapes of the electric charge generation part **10** and the reading part **20** are plate-shaped. The radiation detector **2** has a stacked structure. The electric charge generation part **10** is disposed on the reading part **20**. The electric charge generation part **10** is electrically connected to the reading part **20**. The electric charge generation part **10** generates an electric charge according to the incident radiation. The electric charge generation part **10** outputs the electric charge to the reading part **20**. The reading part **20** generates a pixel signal using the electric charge generated by the electric charge generation part **10**.

(9) The electric charge generation part **10** includes a semiconductor detection part **11**, a control electrode part **12**, and an electric charge output electrode **13**.

(10) The semiconductor detection part **11** generates electron-hole pairs (electric charge pairs) due to the received X-rays. That is, the semiconductor detection part **11** converts the received radiation

into a current signal (an electric charge signal) corresponding to energy thereof. A planar shape of the semiconductor detection part **11** is rectangular. A size of the semiconductor detection part **11** is, for example, about 9.6 mm×9.6 mm. The semiconductor detection part **11** has a semiconductor incident surface **11a** that receives radiation such as X-rays, and a semiconductor output surface **11b** that outputs an electric charge. A signal based on the electric charge is provided from the semiconductor output surface **11b** to the reading part **20**. As the semiconductor detection part **11**, for example, a Cd(Zn)Te electric charge generator, a Si electric charge generator, a Ge electric charge generator, a GaAs electric charge generator, a GaN electric charge generator, a TlBr electric charge generator, or the like may be used. Further, a device including a scintillator and a photodetector may be used as the semiconductor detection part **11**. The scintillator converts X-rays into light. The photodetector converts the light generated by the scintillator into an electric charge.

(11) The control electrode part **12** is provided on the semiconductor incident surface **11a**. The control electrode part **12** covers the entire surface of the semiconductor incident surface **11a**. In a plan view, a planar shape of the control electrode part **12** coincides with the planar shape of the semiconductor incident surface **11a**. The control electrode part **12** forms an electric field in a region in which the semiconductor detection part **11** is disposed. The electric field determines a direction in which the electric charge generated in the semiconductor detection part **11** moves. The control electrode part **12** generates the electric field inside the semiconductor detection part **11** so that the electric charge moves toward the semiconductor output surface **11b**.

(12) A plurality of electric charge output electrodes **13** are provided on the semiconductor output surface **11b**. Each of the electric charge output electrodes **13** may be a bump. One electric charge output electrode **13** corresponds to one pixel G. Therefore, a region in which the electric charge output electrodes **13** are disposed is a pixel region. For example, when n electric charge output electrodes **13** are disposed on a first side and m electric charge output electrodes **13** are disposed on a second side, the total number of electric charge output electrodes **13** is k ($k=n\times m$). In this case, it can be said that the number of pixels of one radiation detector **2** is k. Additionally, when the radiation imaging apparatus **1** has p radiation detectors **2**, it can be said that the number of pixels of the radiation imaging apparatus **1** is r ($r=k\times p$).

(13) The number of pixels of one radiation detector **2** is, for example, 96. Therefore, the number of electric charge output electrodes **13** is also 96. As described above, assuming that the size of the semiconductor detection part **11** is a square of about 9.6 mm×9.6 mm, a distance between the electric charge output electrodes **13** is 100 μm. In FIG. 2 and the like, for convenience of illustration, the number of electric charge output electrodes **13** is shown as a number less than 96.

(14) The reading part **20** generates a pixel signal based on the electric charge generated by the semiconductor detection part **11**. The pixel signal is a digital value. The reading part **20** outputs the pixel signal to the circuit board **3**. The reading part **20** includes an intermediate substrate **30** and a lead-out substrate **40**.

(15) An intermediate substrate **30** guides the electric charge generated by the semiconductor detection part **11** to a lead-out substrate **40**. Although the details will be described later, the arrangement of the electric charge output electrode **13** is different from the arrangement of a first lead-out electrode **41**. Therefore, the intermediate substrate **30** functions to connect the electrodes disposed at different positions from each other. This function is called pitch conversion. Therefore, the intermediate substrate **30** is a pitch conversion substrate. The intermediate substrate **30** converts a pitch of the electric charge output electrode **13** of the semiconductor detection part **11** into a pitch of the first lead-out electrode **41** of the lead-out substrate **40**.

(16) The intermediate substrate **30** has an intermediate input surface **30a** and an intermediate output surface **30b**. The intermediate input surface **30a** faces the electric charge generation part **10**. The intermediate output surface **30b** faces the lead-out substrate **40**.

(17) The intermediate input surface **30a** faces the semiconductor output surface **11b** of the semiconductor detection part **11**. A rectangular intermediate input region **30S** is set on the

intermediate input surface **30a** in a plan view. The intermediate input region **30S** is set over almost the entire surface of the intermediate input surface **30a**. For example, the intermediate input region **30S** may be the entire surface of the intermediate input surface **30a**. In the intermediate input region **30S**, a plurality of first intermediate electrodes **31** are disposed two-dimensionally at equal intervals. The electric charge output electrode **13** of the semiconductor detection part **11** is electrically connected to the first intermediate electrode **31**. Therefore, a pitch of the first intermediate electrodes **31** adjacent to each other is the same as a pitch of the electric charge output electrodes **13**. For example, when the pitch of the electric charge output electrodes **13** is 100 μm , the pitch of the first intermediate electrodes **31** is also 100 μm . On the intermediate input surface **30a**, the intermediate input region **30S** corresponds to the pixel region.

(18) A rectangular intermediate output region **30R** is set on the intermediate output surface **30b** in a plan view. In a plan view, the entire intermediate output region **30R** overlaps the intermediate input region **30S**. That is, an area of the intermediate output region **30R** is smaller than an area of the intermediate input region **30S**. A plurality of second intermediate electrodes **32** (second electrodes) are disposed two-dimensionally in the intermediate output region **30R**.

(19) The first intermediate electrodes **31** (refer to FIG. 2) are disposed in the vicinity of a side surface of the intermediate substrate **30**. On the other hand, the second intermediate electrodes **32** (refer to FIG. 3) are not disposed in the vicinity of the side surface of the intermediate substrate **30**. The second intermediate electrode **32** disposed on the outermost circumference is separated from the side surface of the intermediate substrate **30**. A distance from the second intermediate electrode **32** to the side surface of the intermediate substrate **30** is larger than a distance from the first intermediate electrode **31** to the side surface of the intermediate substrate **30**. For example, the distance from the second intermediate electrode **32** to the side surface of the intermediate substrate **30** is larger than a length of several second intermediate electrodes **32**. This distance corresponds to a width of a lead-out output part **40R** which will be described later. Assuming that the width of the lead-out output part **40R** is 200 μm , the distance from the second intermediate electrode **32** to the side surface of the intermediate substrate **30** is also approximately 200 μm .

(20) As shown in FIG. 4, one first intermediate electrode **31** is electrically connected to one second intermediate electrode **32**. Therefore, the intermediate substrate **30** has a wiring **33** that connects the first intermediate electrode **31** to the second intermediate electrode **32**. A configuration of the wiring **33** is not particularly limited. For example, the wiring **33** may be a combination of a portion that extends in a thickness direction of the intermediate substrate **30** and a portion that extends in a direction of a surface of the intermediate substrate **30**. Further, the wiring **33** may extend linearly from the first intermediate electrode **31** to the second intermediate electrode **32**. Such a wiring structure is referred to as a diagonal via. According to the diagonal via, a pitch conversion distance that is a wiring distance from the first intermediate electrode **31** to the second intermediate electrode **32** can be shortened. As a result, the influence of stray capacitance can be reduced.

(21) The number of the first intermediate electrodes **31** is the same as the number of the second intermediate electrodes **32**. A pitch **P32** of the second intermediate electrodes **32** adjacent to each other is smaller than a pitch **P31** of the first intermediate electrodes **31** adjacent to each other. Assuming that the pitch **P31** of the first intermediate electrode **31** is 100 μm , 80 μm can be exemplified as the pitch **P32** of the second intermediate electrode **32**. In the present embodiment, the plurality of second intermediate electrodes **32** are disposed at equal intervals, but the pitch **P32** of the second intermediate electrodes **32** is not limited to the equal intervals.

(22) Referring to FIGS. 2 and 3 again, the lead-out substrate **40** receives an electric charge and generates a pixel signal corresponding to the received electric charge. The lead-out substrate **40** has a lead-out input surface **40a** and a lead-out output surface **40b**. The lead-out input surface **40a** faces the intermediate output surface **30b**. The lead-out output surface **40b** faces the circuit board **3**. Further, the lead-out substrate **40** includes a lead-out input part **40S**, a lead-out output part **40R**, and a plurality of signal processing parts **45** (refer to FIG. 4).

(23) The lead-out input part **40S** is formed on the lead-out input surface **40a**. The lead-out input part **40S** includes a plurality of first lead-out electrodes **41**. The arrangement of the first lead-out electrodes **41** corresponds to the arrangement of the second intermediate electrodes **32**. That is, the first lead-out electrodes **41** face the second intermediate electrodes **32**. As a result, one first lead-out electrode **41** is electrically connected to one second intermediate electrode **32**. Then, the first lead-out electrode **41** receives an electric charge from the second intermediate electrode **32**.

(24) The lead-out output part **40R** functions as a signal input and output part between the radiation detector **2** and another component (the the control part **5** and the image generation part **4**). For example, the lead-out output part **40R** receives a control signal from the control part **5**. Further, the lead-out output part **40R** outputs a digital value to the image generation part **4**. The lead-out output part **40R** is formed to surround the lead-out input part **40S**. For example, a width of the lead-out output part **40R** is 200 μm . In a plan view, the lead-out output part **40R** has a frame shape.

(25) The lead-out output part **40R** includes a plurality of second lead-out electrodes **42**. For example, in the frame-shaped lead-out output part **40R**, 50 second lead-out electrodes **42** are disposed on one side. Therefore, 200 second lead-out electrodes **42** are disposed on the four sides constituting the lead-out output part **40R**. One second lead-out electrode **42** is connected to one signal processing part **45**. The second lead-out electrode **42** surrounds the lead-out input part **40S**. In other words, the plurality of second lead-out electrodes **42** are disposed on the lead-out input surface **40a** to surround the plurality of first lead-out electrodes **41**. Each of the second lead-out electrode **42** is a through electrode. That is, the second lead-out electrode **42** reaches from the lead-out input surface **40a** to the lead-out output surface **40b**. Each of the second lead-out electrodes **42** is electrically connected to the circuit board **3** via a bump provided on the lead-out output surface **40b**.

(26) As shown in FIG. 4, the signal processing part **45** converts an electric charge into a pixel signal which is the digital value. One signal processing part **45** is connected to one first lead-out electrode **41**. That is, one signal processing part **45** is connected to one pixel G. Therefore, the reading part **20** includes a plurality of signal processing parts **45**. The signal processing part **45** receives an electric charge from the lead-out input part **40S**. The signal processing part **45** outputs a digital value from the lead-out output part **40R**. More specifically, the signal processing part **45** converts the charge output by the electric charge generation part **10** into an energy integration signal which is a pixel signal. The energy integration signal includes at least information on the energy possessed by the incident radiation.

(27) As shown in FIG. 5, the signal processing part **45** includes a signal conversion part **46**, a memory **47**, and a clock signal generation part **48**. Since the signal processing part **45** is connected for each of the pixels G, the number of signal processing parts **45** is the same as the number of pixels G. Similarly, the number of signal conversion parts **46** is the same as the number of pixels G, the number of memories **47** is the same as the number of pixels G, and the number of clock signal generation parts **48** is also the same as the number of pixels G. That is, one signal conversion part **46**, one memory **47**, and one clock signal generation part **48** are connected to one pixel G.

(28) The signal conversion part **46** is connected to the electric charge output electrode **13** of the electric charge generation part **10** via the wiring **33** of the intermediate substrate **30**. The signal conversion part **46** receives an electric charge $\phi 1$ from the electric charge generation part **10**. The signal conversion part **46** discretizes an analog signal based on the electric charge $\phi 1$. The analog signal is expressed as a voltage. The voltage corresponds to the energy of radiation or the number of particles incident on the corresponding pixel G of the electric charge generation part **10**. Therefore, the signal conversion part **46** is an AID converter that outputs a digital signal. For example, a resolution of the signal conversion part **46** may be 10 bits.

(29) The signal conversion part **46** is operated by a clock signal $\theta 2$ provided by the clock signal generation part **48**. When the signal conversion part **46** receives the clock signal $\theta 2$, the signal conversion part **46** performs an operation of transmitting an analog signal generated due to the

incidence of radiation from the pixel G, an operation of resetting the electric charge accumulated in the pixel G, and an operation of converting the transmitted analog signal into a digital signal. In addition to the above-described operations, the signal conversion part **46** may perform other desired operations.

(30) The clock signal generation part **48** generates the clock signal $\theta 2$ necessary for the operation of the signal conversion part **46**. Then, the clock signal generation part **48** provides the clock signal $\theta 2$ to the signal conversion part **46**. That is, one clock signal generation part **48** is connected to one signal conversion part **46**. In other words, an output destination of a certain clock signal generation part **48** is one specified signal conversion part **46**. The signal conversion part **46** is provided for each of the pixels G. Therefore, it can be said that the clock signal generation part **48** is also provided for each of the pixels G. With such a configuration, the clock signal generation part **48** may be disposed to be adjacent to the signal conversion part **46**. As a result, a length of a signal line from the clock signal generation part **48** to the signal conversion part **46** can be shortened. When the signal line is shortened, a transmission distance of the clock signal $\theta 2$ is also shortened, and thus signal delay is less likely to occur. Further, a frequency of the clock signal can be increased to about 100 MHz.

(31) The clock signal generation part **48** starts generation of the clock signal $\theta 2$ triggered by the incidence of radiation on the corresponding pixel G (refer to a reference numeral **S1** in FIG. **6**). The clock signal generation part **48** outputs a predetermined number of pulses in accordance with the trigger. The number of output pulses may be the number required for an analog-to-digital conversion operation in the signal conversion part **46**. For example, when one incidence of radiation is detected, the clock signal generation part **48** outputs **16** pulses as the clock signal $\theta 2$.

(32) Since the clock signal generation part **48** is provided for each of the pixels G, the operations related to each of the pixels G can be made independent. In other words, it is not necessary to synchronize the operations of the plurality of signal conversion parts **46**. Therefore, a signal for synchronizing the operations of the plurality of signal conversion parts **46** is also unnecessary. For example, when radiation is incident on a certain pixel G, the signal conversion part **46** can immediately start an operation of generating a digital signal. That is, it can be said that the operation of the signal conversion part **46** is synchronized with the incidence of radiation. According to such an operation, since a signal conversion process is performed immediately after the radiation is incident, a problem of phase shift does not occur. In addition, no unnecessary waiting time is generated. Such an operation has a high affinity for a photon counting type detector that performs the signal conversion process for each randomly incident particle.

(33) Further, even when radiation is incident on the plurality of pixels G at a certain timing, the operation of independently generating the digital signals can be started. Further, it is also possible to operate only the signal conversion part **46** corresponding to the pixel G on which the radiation is incident, and not to operate the signal conversion part **46** corresponding to the pixel G on which the radiation is not incident. When the signal conversion operation is uniformly performed for all the pixels G, the power consumption increases in proportion to the number of pixels as the number of pixels increases. However, since the radiation imaging apparatus **1** of the present embodiment can operate only the necessary signal conversion part **46**, the power consumption is not proportional to the number of pixels. Therefore, it is possible to achieve both an increase in the number of pixels and suppression of an increase in the power consumption.

(34) The memory **47** is connected to the signal conversion part **46**. The memory **47** receives a digital signal $\phi 2$ from the signal conversion part **46**. Then, the memory **47** stores the digital signal $\phi 2$ each time when the digital signal ($\phi 2$) is input. The memory **47** sequentially stores the digital signal ($\phi 2$) in a predetermined memory space. The memory **47** is connected to the control part **5** via a wiring **51** (a wiring part). An input of the wiring **51** is one, and an output of the wiring **51** is a plurality. Additionally, the input of the wiring **51** is connected to the control part **5**, and the output of the wiring **51** is connected to the memory **47** of the signal processing part **45**. That is, a

transmission signal $\theta 1$ is output from one control part 5 to the plurality of memories 47. The control part 5 that outputs the transmission signal $\theta 1$ has a function as a transmission signal generation part.

(35) The memory 47 outputs a digital signal $\phi 2$ to the image generation part 4 in accordance with the transmission signal $\theta 1$ provided by the control part 5. That is, since the memory 47 does not receive the clock signal $\theta 2$, it is not controlled by the clock signal $\theta 2$. In other words, the operation of the signal conversion part 46 and the operation of the memory 47 are based on signals different from each other. Therefore, each of the operations is also independent of each other.

(36) The radiation imaging apparatus 1 includes the plurality of pixels G that generate an electric charge corresponding to energy or the number of particles of the incident radiation, the plurality of signal processing parts 45 that are respectively connected to the plurality of pixels G and generate a digital value based on the electric charge provided by each of the pixels G, and the circuit board 3 in which a radiation detector 2 including the plurality of pixels G and the plurality of signal processing parts 45 is disposed in a two-dimensional manner. Each of the plurality of signal processing parts 45 includes the signal conversion part 46 that converts an analog value based on the electric charge into a digital value, and the clock signal generation part 48 that provides a clock signal for generating a digital value to the signal conversion part.

(37) In the radiation imaging apparatus 1, the signal processing part 45 is provided for each of the pixels G. The signal processing part 45 includes the signal conversion part 46 and the clock signal generation part 48. That is, in the radiation imaging apparatus 1, the clock signal generation part 48 is provided for each of the pixels G. With such a configuration, it is possible to drive only the signal conversion part 46 corresponding to the pixel G on which the radiation is incident, and not to drive the signal conversion part 46 corresponding to the pixel G on which the radiation is not incident. Therefore, even when the number of pixels G increases, the signal conversion part 46 which does not need to be operated is not operated, and thus an increase in the power consumption can be suppressed.

(38) The radiation imaging apparatus 1 of the present invention is not limited to the above-described embodiment.

REFERENCE SIGNS LIST

(39) 1 . . . Radiation imaging apparatus, 2 . . . Radiation detector, 3 . . . Circuit board, 4 . . . Image generation part, 5 . . . Control part, 10 . . . Electric charge generation part, 20 . . . Reading part, 11 . . . Semiconductor detection part, 12 . . . Control electrode part, 13 . . . Electric charge output electrode, 11a . . . Semiconductor incident surface, 11b . . . Semiconductor output surface, 30 . . . Intermediate substrate, 40 . . . Lead-out substrate, 30a . . . Intermediate input surface (first main surface), 30b . . . Intermediate output surface (second main surface), 30S . . . Intermediate input region, 31 . . . First intermediate electrode (first electrode), 30R . . . Intermediate output region, 32 . . . Second intermediate electrode (second electrode), 40a . . . Lead-out input surface, 40b . . . Lead-out output surface, 41 . . . First lead-out electrode, 42 . . . Second lead-out electrode, 45 . . . Signal processing part, 46 . . . Signal conversion part, 47 . . . Memory, 48 . . . Clock signal generation part.

Claims

1. A radiation imaging apparatus comprising: a plurality of pixels configured to generate an electric charge corresponding to energy or the number of particles of incident radiation; a plurality of signal processing parts respectively connected to the plurality of pixels and configured to generate a digital value based on the electric charge provided by each of the pixels; and a circuit board in which a radiation detector including the plurality of pixels and the plurality of signal processing parts is disposed in a two-dimensional manner, wherein each of the plurality of signal processing parts includes a signal conversion part configured to convert an analog value based on the electric charge into the digital value, and a clock signal generation part configured to provide the signal

conversion part with a clock signal to convert the analog value based on the electric charge into the digital value.

2. The radiation imaging apparatus according to claim 1, wherein the clock signal generation part starts an operation of providing the clock signal to the signal conversion part triggered by generation of the electric charge due to incidence of the radiation on the pixels, and wherein a number of output pulses of the clock signal is a predetermined number required for a conversion operation of the signal conversion part.

3. The radiation imaging apparatus according to claim 1, further comprising: a transmission signal generation part configured to provide a transmission signal for outputting the digital value from the signal processing part to the circuit board to each of the plurality of signal processing parts; and a wiring unit configured to transmit the transmission signal to the signal processing part.

4. The radiation imaging apparatus according to claim 2, further comprising: a transmission signal generation part configured to provide a transmission signal for outputting the digital value from the signal processing part to the circuit board to each of the plurality of signal processing parts; and a wiring unit configured to transmit the transmission signal to the signal processing part.
